

Loops in JavaScript

Loops allow you to execute a block of code multiple times, which is useful for tasks like iterating over arrays or repeating operations until a condition changes.

1. `for` Loop

Syntax

```
for (initialization; condition; finalExpression) {  
    // code to execute on each iteration  
}
```

Example

```
for (let i = 0; i < 5; i++) {  
    console.log(i); // prints 0, 1, 2, 3, 4  
}
```

- **initialization**: executed once before the loop starts (e.g., `let i = 0`)
 - **condition**: checked before each iteration; if `true` , the loop continues
 - **finalExpression**: executed at the end of each iteration (e.g., `i++`)
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2. `while` Loop

Syntax

```
while (condition) {  
    // code to execute while condition is true  
}
```

Example

```
let count = 0;  
while (count < 3) {  
    console.log(count); // prints 0, 1, 2  
    count++;  
}
```

- The condition is evaluated before each iteration.
 - If the condition is initially `false`, the loop body may never run.
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3. do...while Loop

Syntax

```
do {  
    // code to execute  
} while (condition);
```

Example

```
let num = 0;  
do {  
    console.log(num); // prints 0  
    num++;  
} while (num < 1);
```

- The loop body executes **at least once**, then the condition is checked.
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4. `for...of` Loop

Purpose: Iterate over iterable objects (arrays, strings, etc.).

Syntax

```
for (variable of iterable) {  
    // code using variable  
}
```

Example

```
let colors = ["red", "green", "blue"];  
for (let color of colors) {  
    console.log(color);  
}
```

5. `for...in` Loop

Purpose: Iterate over the keys of an object.

Syntax

```
for (key in object) {  
    // code using key and object[key]  
}
```

Example

```
let person = { name: "Alice", age: 30 };  
for (let prop in person) {  
    console.log(prop + ": " + person[prop]);  
    // prints "name: Alice" and then "age: 30"  
}
```

Summary

- `for` : general-purpose loop with counter.
- `while` : loop based on a condition.
- `do...while` : condition checked after first execution.
- `for...of` : iterate over iterable values.
- `for...in` : iterate over object keys.